

TECHBIG SOLUTIONS

SAP S/4HANA SECURITY – End-to-End

Complete Professional Training Syllabus

Ambattur, Chennai

Course Objective

A complete end-to-end SAP S/4HANA Security curriculum covering security architecture, users, roles, authorization objects, PFCG, Fiori security, Gateway/OData, CDS access controls, HANA security, RFC/technical users, identity and access management, SSO, audit/compliance, SoD, troubleshooting, transports, cloud identity concepts and real-time implementation/support scenarios.

SAP's current security learning path covers authorization concepts, user access, SAP Identity Access Management, HANA user administration, S/4HANA role maintenance, Fiori authorizations, Cloud Identity Services and authorization troubleshooting. SAP's Fiori security curriculum also includes business catalogs, business roles, spaces/pages, CDS authorization and analysis of missing authorizations. ■cite■turn0search0■turn0search1■turn0search14■

1. SAP S/4HANA Security Fundamentals

- SAP security objectives and principles
- S/4HANA security architecture
- ABAP application-server security overview
- Authentication versus authorization
- Users, roles, profiles and authorization objects
- Least privilege and need-to-know
- Security in DEV/QAS/PRD landscapes
- Security governance and operational responsibilities

2. SAP User Administration

- SU01 user maintenance
- User types and technical users
- User groups
- Password and logon parameters
- Validity and account locking
- Mass user administration
- SU10 mass maintenance
- Reference users and service users
- Communication and system users
- User master troubleshooting

3. Authorization Concepts

- Authorization objects
- Authorization fields
- Authorization classes
- Authorization object values
- Start authorizations
- Data access authorizations
- Organizational levels
- Composite authorization concepts
- Authorization defaults
- Critical authorization principles

4. PFCG Role Administration

- Single roles
- Composite roles
- Derived roles
- Role menu
- Transactions

- Authorization data
- Organizational levels
- Generate profiles
- User assignment
- Mass role assignment
- Role comparison
- Role transport
- Role documentation

5. SU24 & Authorization Proposal Management

- SU24 fundamentals
- Transaction-to-authorization proposals
- Check indicators
- Maintaining authorization defaults
- Impact of SU24 on PFCG
- Custom transaction authorization proposals
- Troubleshooting missing authorization objects
- Upgrade and support-package considerations

6. Authorization Troubleshooting

- SU53 analysis
- STAUTHTRACE / authorization trace
- ST01 trace concepts
- Missing authorization diagnosis
- Failed transaction analysis
- RFC authorization errors
- OData authorization errors
- Fiori authorization errors
- Comparing working and non-working users
- Root-cause analysis and evidence collection

7. SAP Fiori Security & Authorization

- Fiori architecture
- Fiori Launchpad security
- Fiori apps and authorization requirements
- Business catalogs
- Business roles
- Spaces and pages
- Target mappings
- Launchpad content assignment
- Frontend/backend authorization concepts

- OData authorizations
- Fiori rapid activation overview
- Analyzing missing Fiori authorizations

8. Fiori Role Design – End-to-End

- Identify business process and personas
- Select SAP-delivered business roles
- Copy/refine standard roles
- Create custom business catalogs
- Assign catalogs to roles
- Configure spaces and pages
- Assign users
- Test application access
- Analyze missing permissions
- Transport Fiori authorization content

9. SAP Gateway & OData Security

- Gateway architecture
- IWFND and IWBEP concepts
- OData service authorization
- SICF service activation
- Service registration
- System aliases
- Authorization objects for OData
- Error logs and tracing
- Backend authorization dependencies
- Secure OData exposure

10. CDS Access Control & DCL

- ABAP CDS security concept
- CDS access control
- DCL and DEFINE ROLE
- Literal conditions
- PFCG conditions
- Implicit CDS authorization checks
- CDS authorization versus classic AUTHORITY-CHECK
- Field/value-based restrictions
- Testing CDS restrictions
- CDS authorization troubleshooting

11. SAP HANA Security

- HANA security architecture
- HANA users
- HANA roles
- System and object privileges
- Analytic privileges overview
- User administration
- Authentication methods
- Password policies
- HANA auditing
- Database security monitoring
- HANA authorization troubleshooting

12. Identity & Access Management (IAM)

- IAM concepts
- Identity lifecycle
- Authentication and authorization
- User provisioning
- Identity providers
- SAP Cloud Identity Services overview
- Identity Authentication
- Identity Provisioning
- Business users and business roles
- Role-based access management
- Integration with S/4HANA

13. Single Sign-On (SSO)

- SSO architecture
- SAML 2.0 overview
- Kerberos/SPNEGO overview
- X.509 certificate concepts
- Trust configuration
- Identity provider integration
- SSO troubleshooting
- Certificate expiry and renewal
- Browser/login troubleshooting

14. RFC & Technical User Security

- RFC destinations
- SM59 security
- Trusted RFC concepts
- Technical communication users

- RFC authorization objects
- Destination security
- Background technical users
- Interface user governance
- Hardening RFC access
- Troubleshooting RFC failures

15. Security for Interfaces & Integration

- IDoc security
- Web services security
- OData/API security
- RFC security
- HTTP destinations
- TLS/SSL overview
- Certificate management
- Integration technical users
- Secure credential management
- Interface monitoring

16. SAP S/4HANA Cloud Security Overview

- Public Cloud IAM concepts
- Business catalogs
- Business role templates
- Business users
- Restrictions
- IAM apps
- Identity provider integration
- Cloud Identity Services
- Role-based access management
- Cloud security administration

17. Segregation of Duties (SoD)

- SoD principles
- Critical business functions
- Conflicting authorizations
- Risk identification
- Role design for SoD
- Preventive versus detective controls
- User-level risk analysis
- Role-level risk analysis
- Mitigation controls

- Audit evidence

18. SAP Audit & Compliance

- Security audit fundamentals
- Audit log concepts
- SM19/SM20 overview where applicable
- Security Audit Log
- Change documents
- User and role change tracking
- Table logging considerations
- Emergency access concepts
- Compliance evidence
- Internal and external audit support

19. Emergency Access Management (EAM)

- Firefighter concept
- Emergency access principles
- Temporary elevated access
- Log review
- Controller responsibilities
- Approval and workflow concepts
- Audit trail
- Risk and governance
- Integration with SAP GRC overview

20. SAP GRC Access Control Overview

- SAP GRC architecture
- Access Risk Analysis
- Access Request Management
- Business Role Management
- Emergency Access Management
- Risk analysis and mitigation
- Workflow concepts
- Connector concepts
- Synchronization
- Security-role governance

21. SAP Security in S/4HANA Business Processes

- Finance security
- Procurement/MM security
- Sales/SD security

- Production/PP security
- Warehouse/EWM security
- Quality/QM security
- Plant maintenance/PM security
- Business-process authorization matrix
- Org-level design
- End-to-end business-role testing

22. Security Testing & Validation

- Unit testing of roles
- Negative testing
- Positive testing
- Business-process testing
- Fiori app testing
- OData testing
- CDS authorization testing
- Regression testing
- Role comparison
- Authorization trace validation
- User acceptance security testing

23. Role Design & Implementation Methodology

- Requirement gathering
- Business-role mapping
- Authorization matrix
- Org-level determination
- Role naming conventions
- Single/derived/composite strategy
- Least-privilege design
- Build and test cycle
- Transport strategy
- Production cutover
- Post-go-live review

24. Security Transport Management

- Role transport concepts
- Authorization data transport
- Fiori content transport
- CDS/security-related transport
- DEV to QAS to PRD flow
- Transport dependencies

- Post-import validation
- Emergency changes
- Change documentation

25. Security Hardening

- Secure parameter configuration
- Password and logon hardening
- Inactive user cleanup
- Privileged-user governance
- Technical-user restrictions
- RFC hardening
- Gateway/OData hardening
- TLS and certificate management
- Audit-log protection
- Patch and SAP Security Note awareness

26. Security Monitoring & Daily Operations

- Daily user/security checks
- Failed logon monitoring
- Locked-user monitoring
- Critical role changes
- Authorization failures
- Audit log monitoring
- RFC/security incidents
- Fiori access issues
- Certificate monitoring
- Security incident escalation

27. SAP Security Notes & Patch Management

- SAP Security Notes concept
- Security Note analysis
- Prioritization and risk assessment
- Test-before-production approach
- Support Package considerations
- Kernel/security patch coordination
- Vulnerability remediation
- Evidence and documentation

28. Real-Time E2E Security Projects

- Project 1: Build complete S/4HANA authorization matrix
- Project 2: Create MM business role with organizational restrictions

- Project 3: Create SD business role and test SoD
- Project 4: Configure Fiori business catalog, role, space and page
- Project 5: Troubleshoot missing Fiori authorization
- Project 6: Configure CDS access control using DCL/PFCG conditions
- Project 7: Configure and test RFC technical-user access
- Project 8: Configure SSO/identity integration scenario
- Project 9: Perform SoD risk analysis and mitigation
- Project 10: Production security incident investigation and audit evidence

29. Production Support & Troubleshooting

- User cannot log in
- User locked or expired
- Transaction authorization failure
- Missing authorization object
- Fiori tile missing
- Fiori app opens but business data is missing
- OData 401/403 errors
- CDS data restriction issue
- RFC authorization failure
- SSO/certificate issue
- Role transport issue
- Emergency production access
- Security incident documentation and RCA

30. Interview & Certification Preparation

- S/4HANA Security architecture questions
- PFCG/SU24/SU01 questions
- Fiori authorization scenarios
- CDS/DCL authorization questions
- HANA security questions
- IAM and SSO scenarios
- GRC/SoD questions
- Real-time production support scenarios
- Security implementation case studies
- Mock interviews and practical assessment

Recommended Training Roadmap

Phase	Coverage	Outcome
1	Security Foundation	Understand S/4HANA security architecture and concepts
2	Users & PFCG	Build and manage secure authorization roles
3	Fiori & Gateway	Secure Fiori apps, catalogs, OData and Launchpad
4	CDS & HANA	Implement data-level authorization and HANA security
5	IAM & SSO	Manage identity, provisioning and authentication
6	GRC & Compliance	SoD, EAM, audit and governance
7	Implementation	Design, build, test and transport security roles
8	Production Support	Troubleshoot, monitor and perform RCA
9	Projects	Complete E2E S/4HANA security implementation

SAP Reference Alignment

This syllabus is aligned with SAP's current S/4HANA security learning scope, including user access, IAM, HANA user administration, S/4HANA role maintenance, Fiori authorizations and troubleshooting. SAP's Fiori security curriculum explicitly includes business catalogs, business roles, spaces/pages, CDS authorization, missing-authorization analysis and authorization transport. ■cite■turn0search0■turn0search1■turn0search14■

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